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Chiussi et al.

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(54) **UNICASTING AND MULTICASTING
MULTIMEDIA SERVICES**

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H04L 65/1016 (2013.01); *H04L 65/611*
(2022.05)

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See application file for complete search history.

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H04W 4/08 (2009.01)

H04W 8/18 (2009.01)

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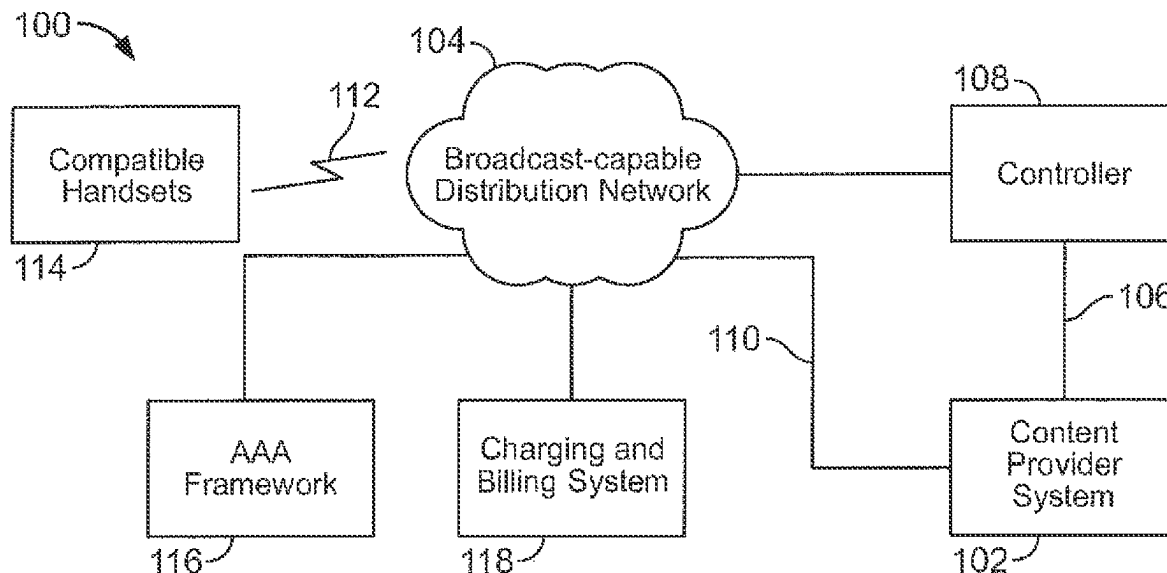
(52) **U.S. Cl.**

CPC *H04W 4/08* (2013.01); *H04L 12/189*
(2013.01); *H04N 21/6405* (2013.01); *H04N*

(57) **ABSTRACT**

A network entity controls delivery of information to user devices over bearer paths including unicast channels and multicast channels. The network entity may interoperate in any of a number of network architectures, including 3GPP Internet Protocol Multimedia Subsystem (IMS) and 3GPP2 Multimedia Domain (MMD). The network entity may provide functionality of a modified 3GPP2 Broadcast and Multicast Service (BCMCS) controller component configured to enable BCMCS signaling protocol transactions to occur over 3GPP IMS interfaces and/or 3GPP2 MMD interfaces. A network entity configured to interoperate in a 3GPP IMS and/or 3GPP2 MMD network architecture may provide network-mobile multimedia services to user devices. Content associated with the multimedia services may be stored in storage devices in the network. A common interface through which a network operator defines service-specific parameters of a number of unicast and multicast multimedia services deployed in a distribution network may be provided.

22 Claims, 10 Drawing Sheets



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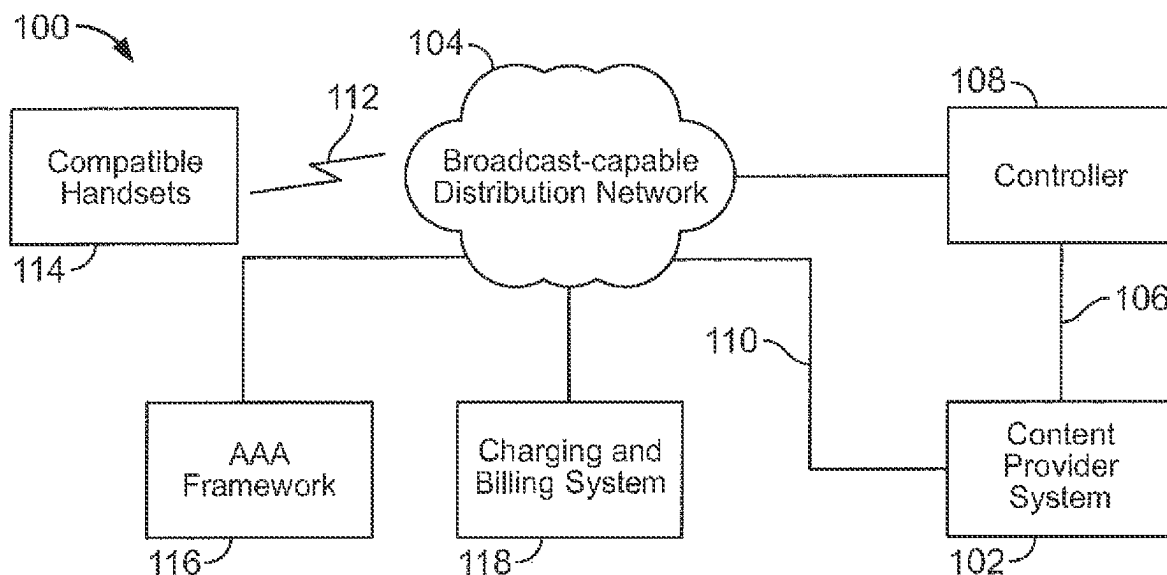


FIG. 1

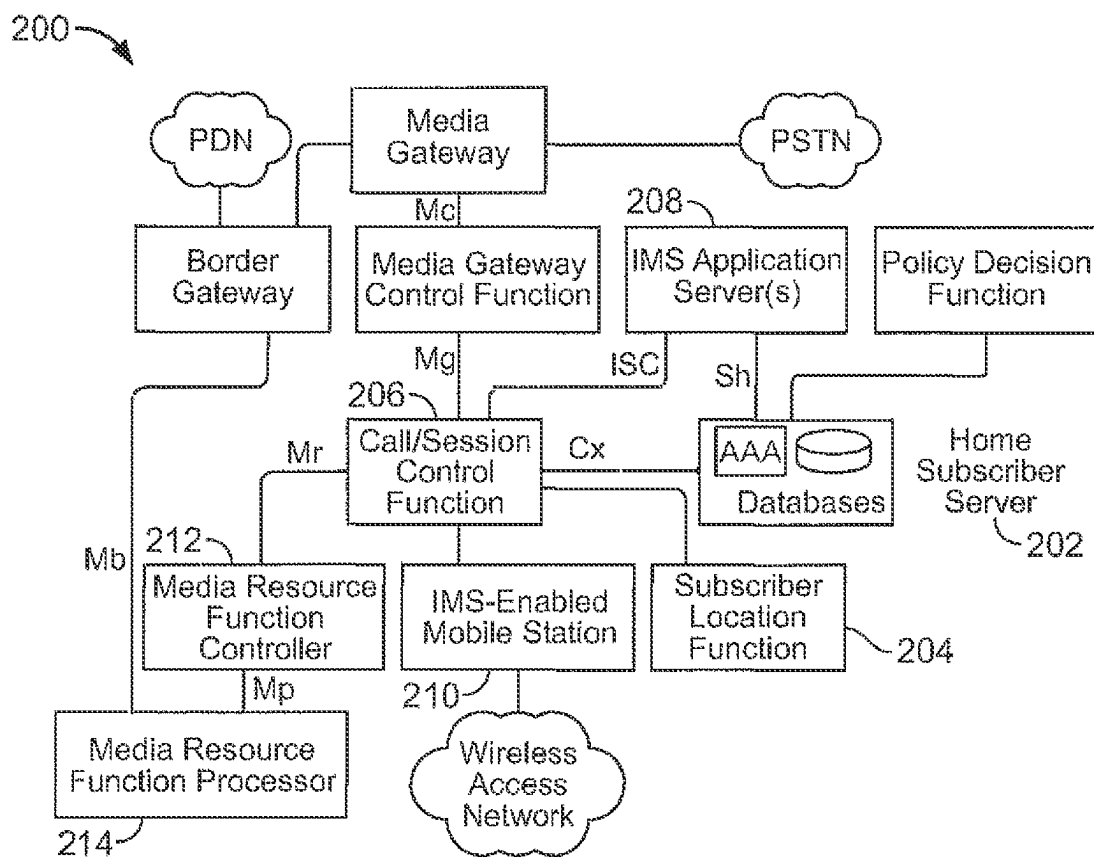


FIG. 2

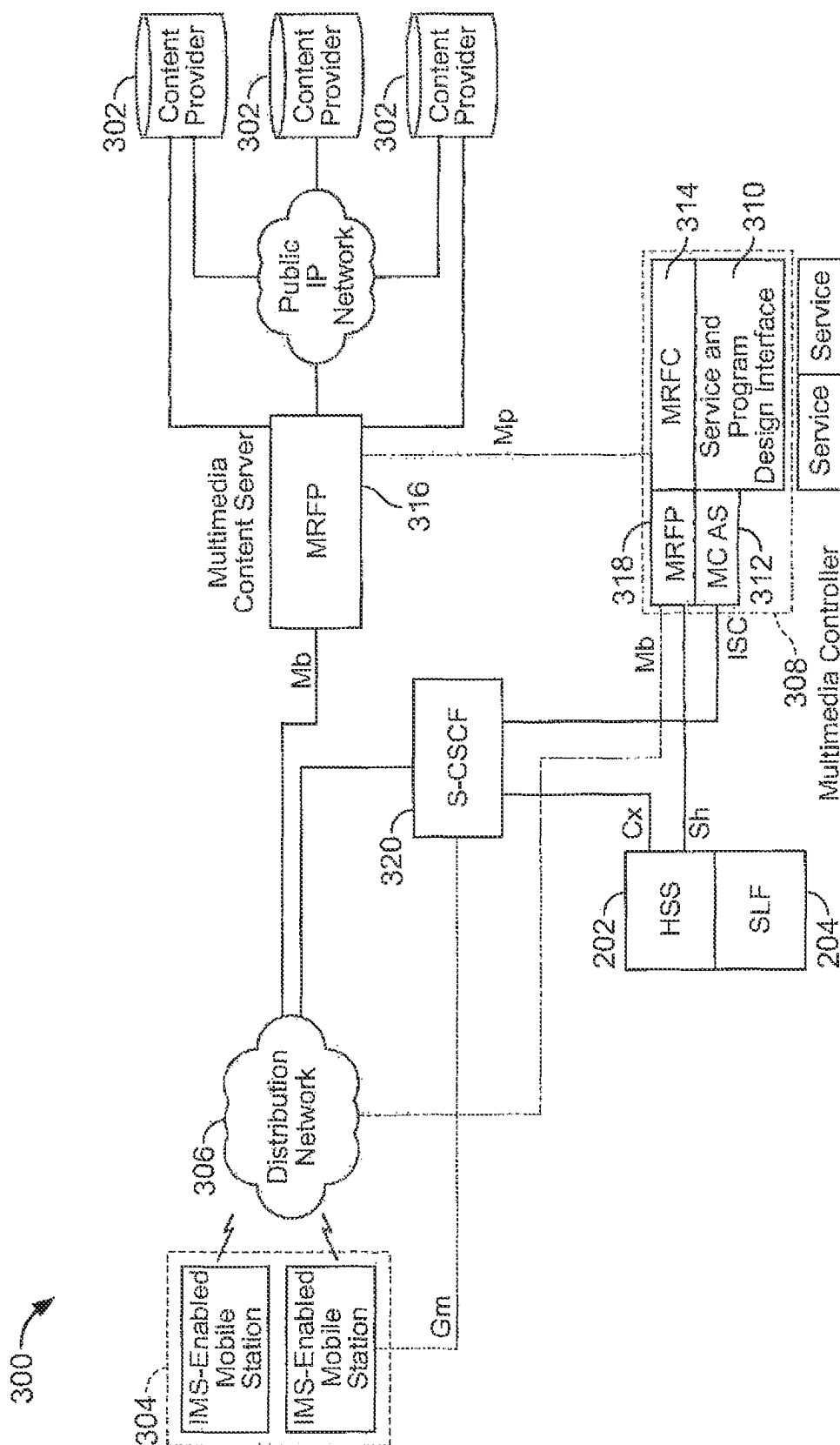


FIG. 3

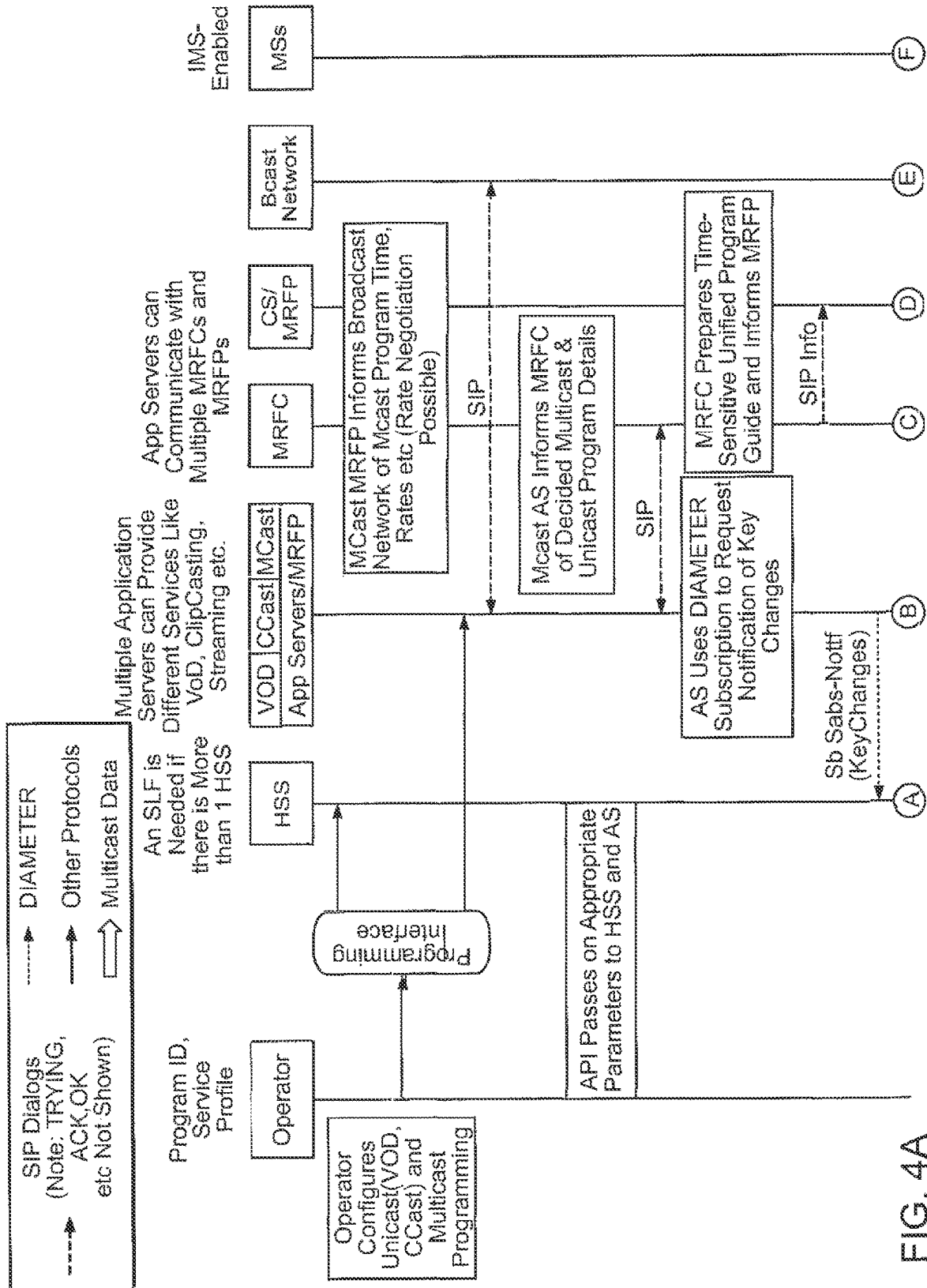


FIG. 4A

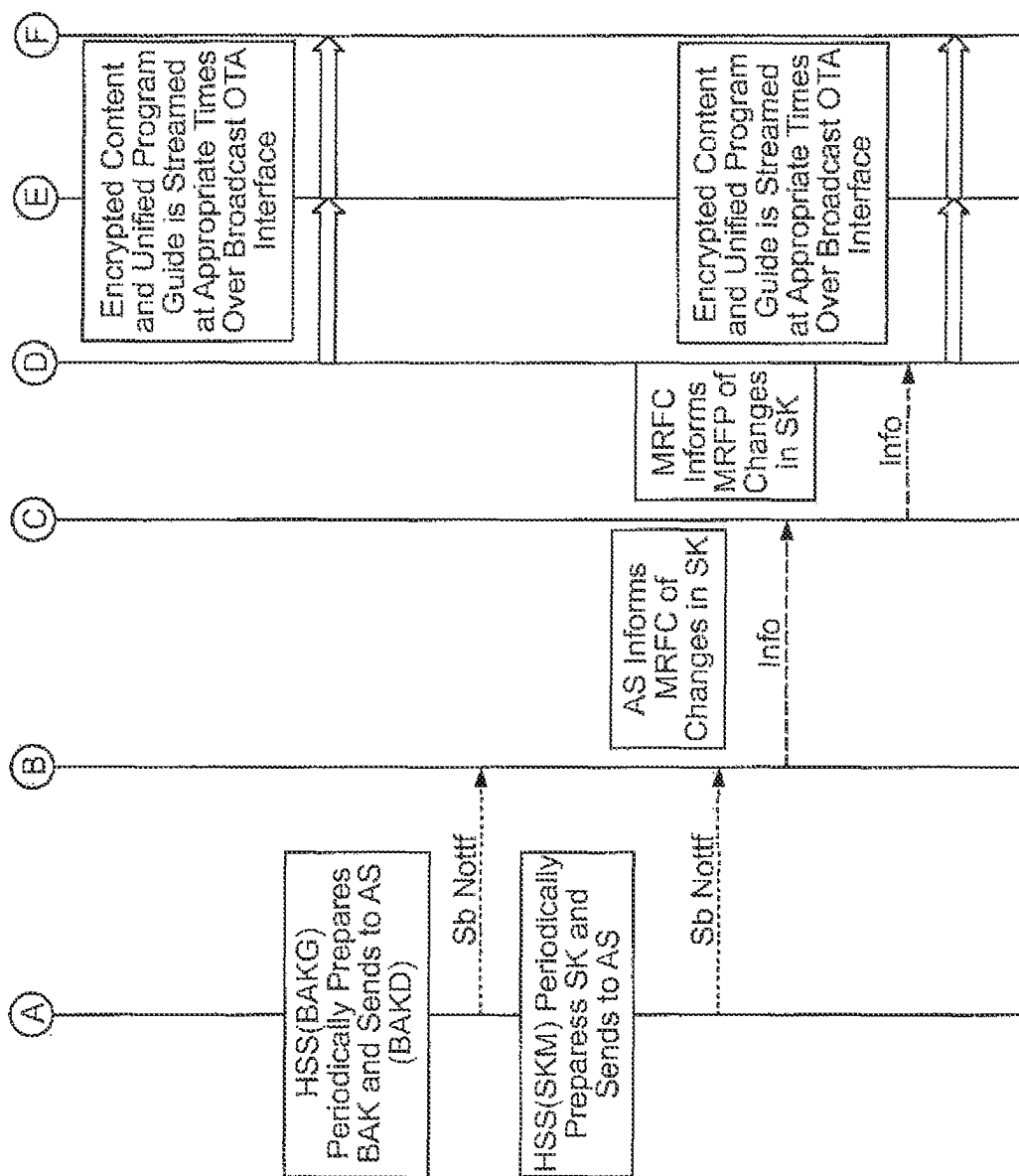
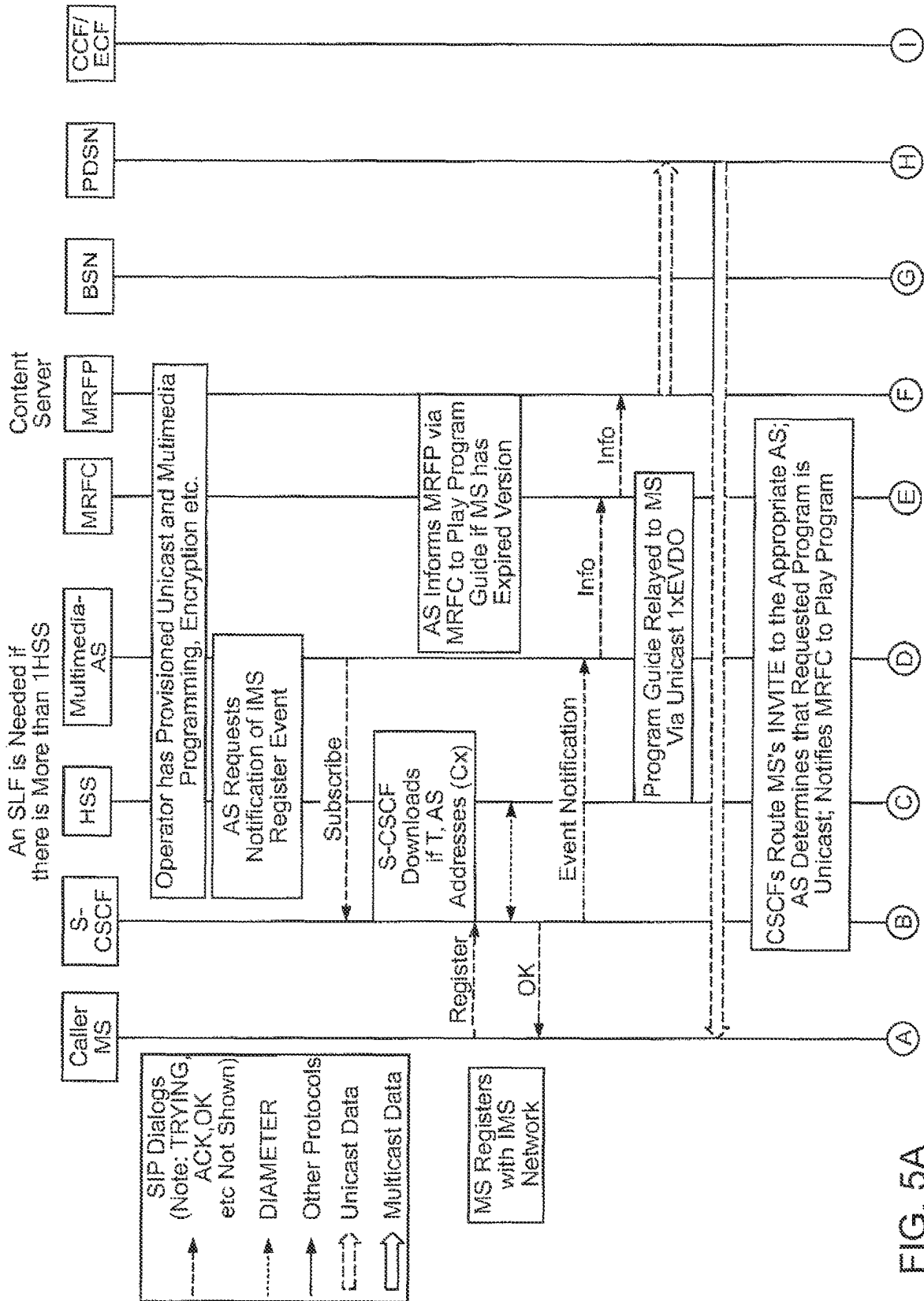


FIG. 4B



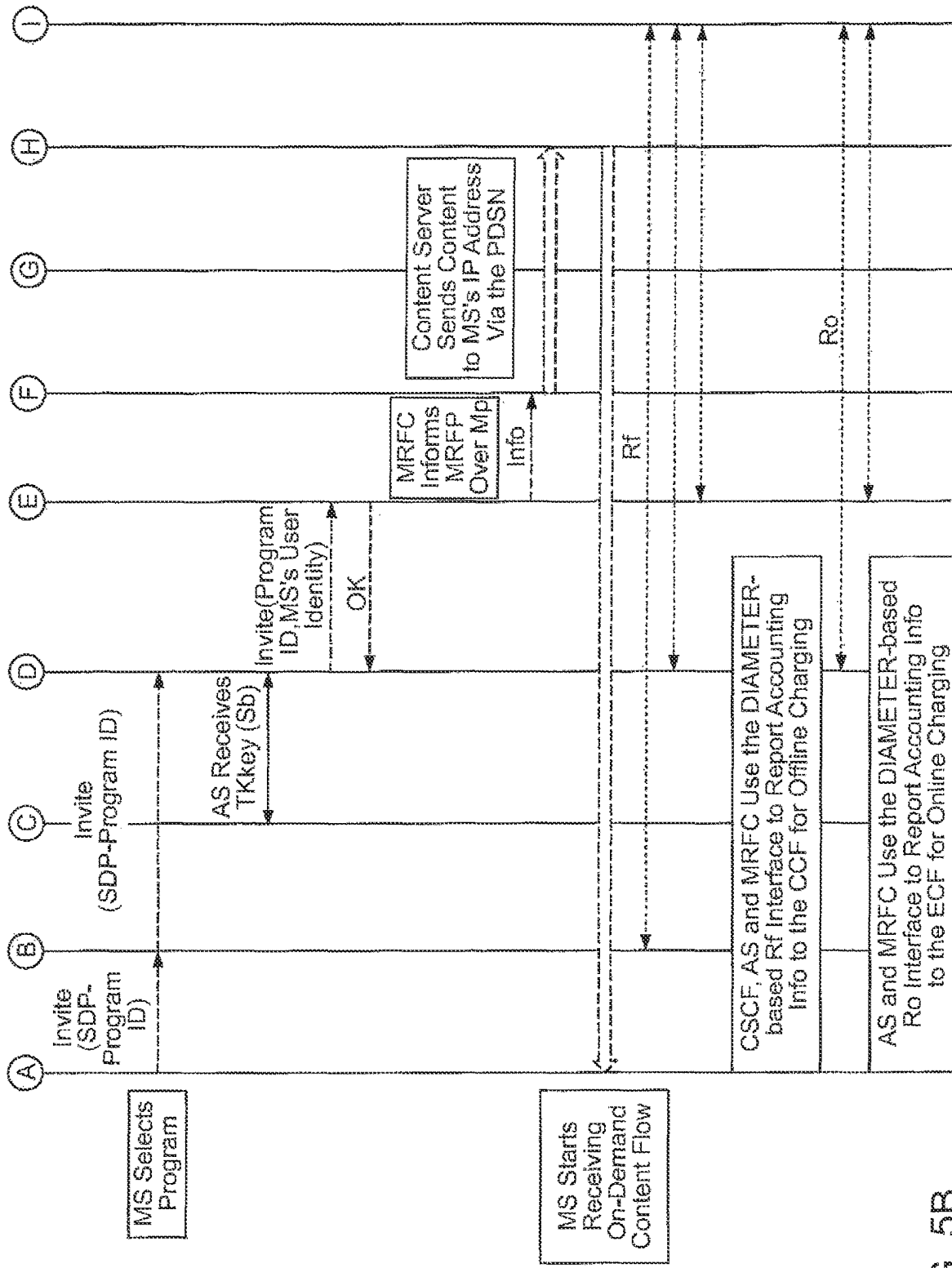
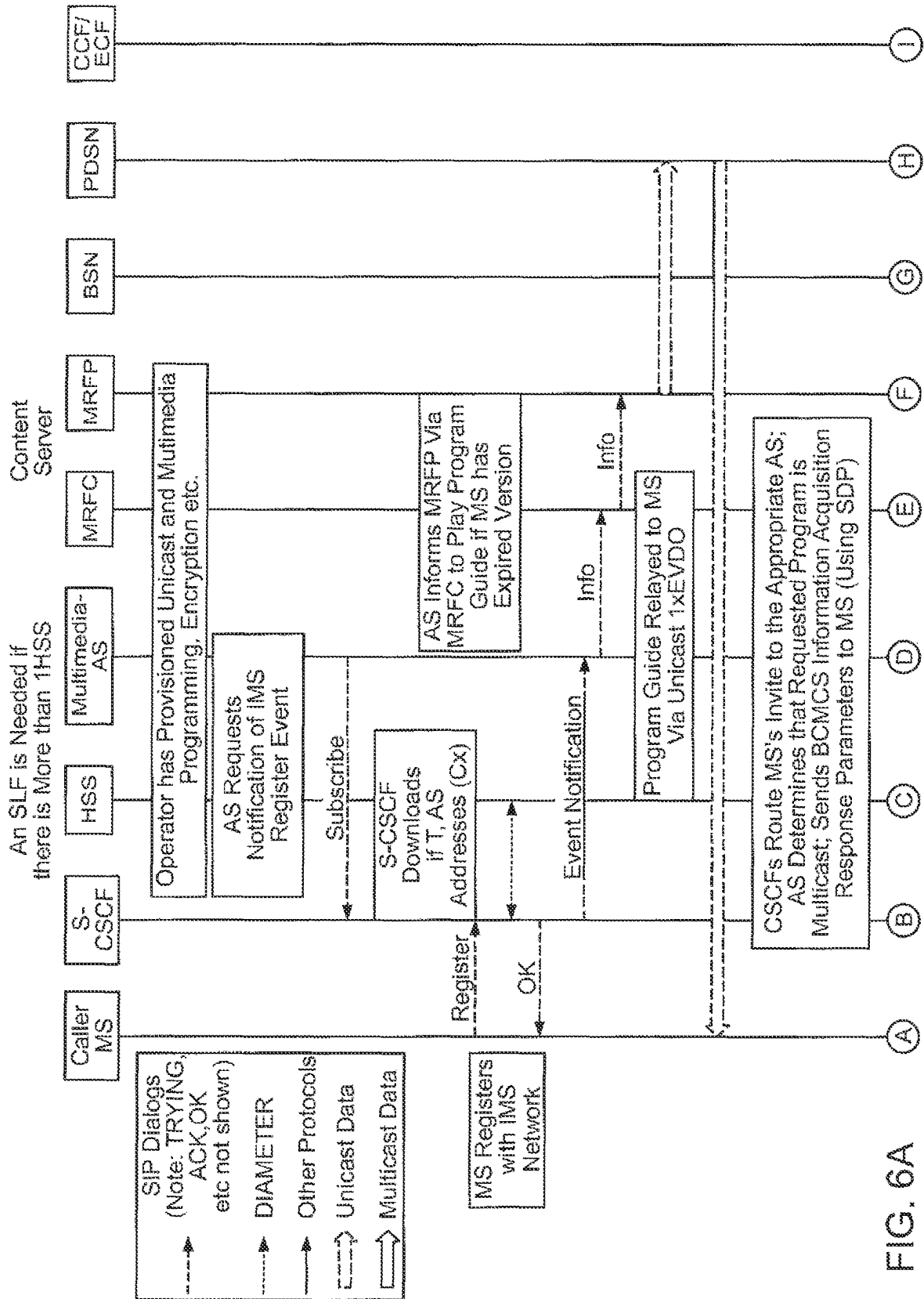


FIG. 5B



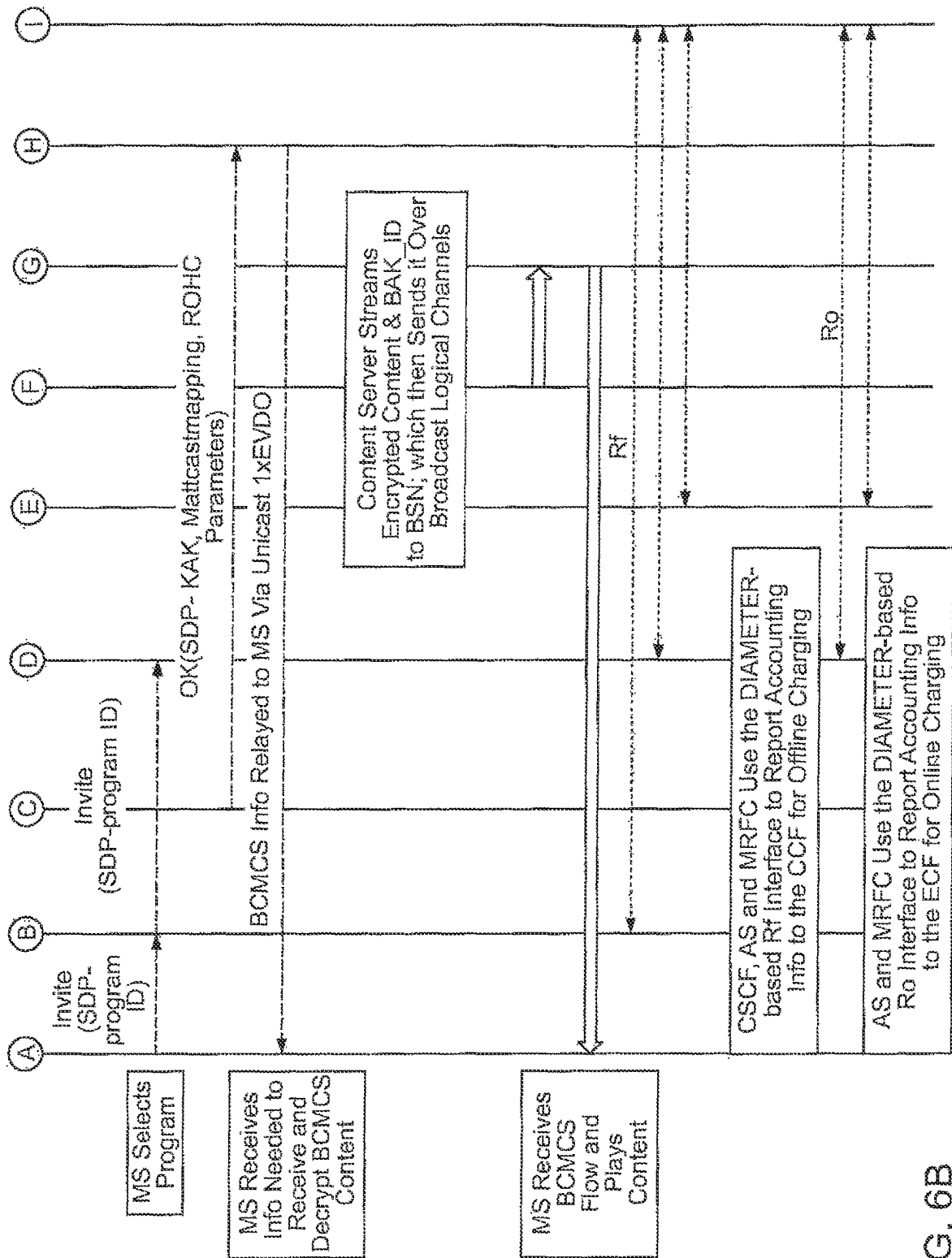


FIG. 6B

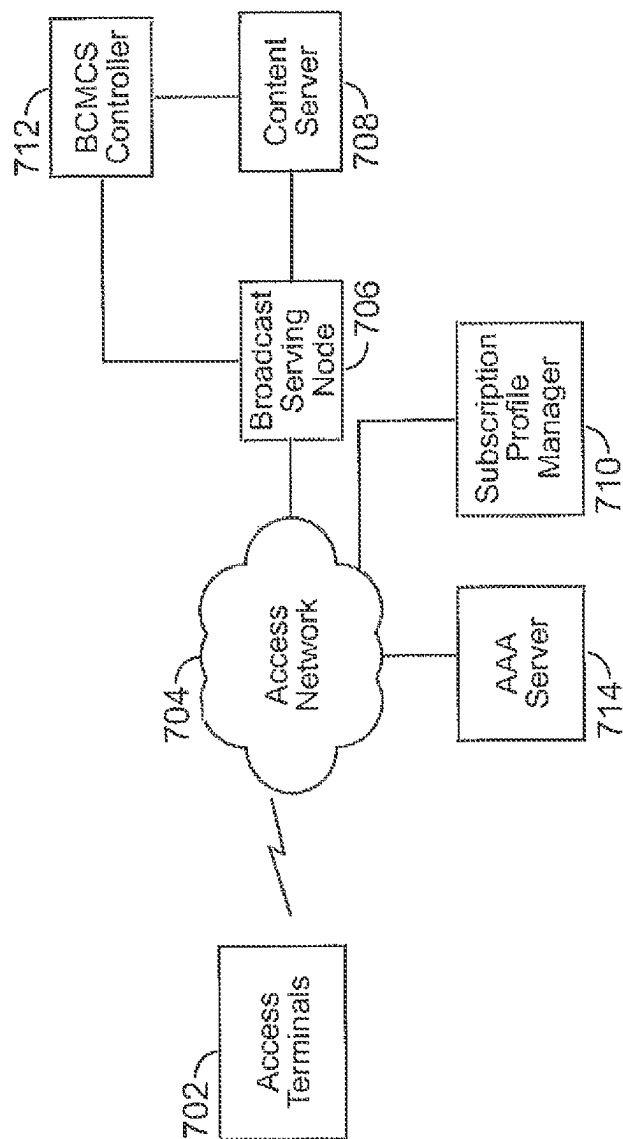


FIG. 7

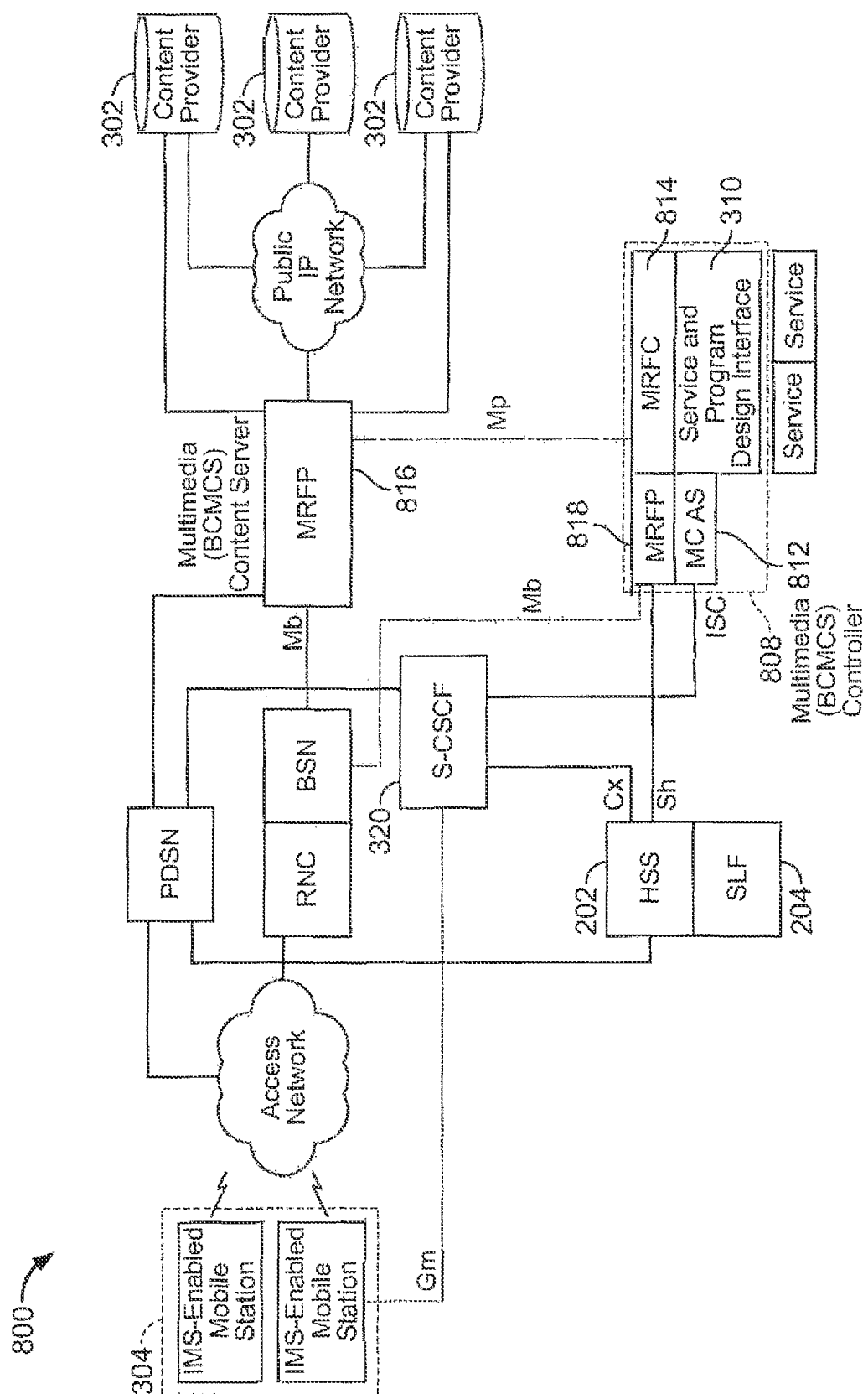


FIG. 8

1

UNICASTING AND MULTICASTING MULTIMEDIA SERVICES

BACKGROUND

This description relates to multicasting and unicast multimedia services.

Streaming movies, sports-clips, weather forecasts, news-casts, music, etc. have become increasingly popular with users. Multimedia streams—combinations of audio, visual, and data formats—convey a richness of content that enhances user experience. New and innovative multimedia services, as well as variations of existing multimedia services, are generated on a continuing basis. Examples of multimedia services include television and video services, video on demand (VOD), private video recording (PVR), radio, music distribution services, clip casting, pod casting, push-to-talk (PTT) services, push-to-media (PTM) services, multimedia messaging services (MMS), interactive gaming, multimedia conferencing, video telephony, and venue casting.

Initially, these services were delivered exclusively over high-speed wireline or cable networks. Over time, demand has rapidly grown for delivery of multimedia services through an air interface to a variety of mobile devices, such as cell phones, personal digital assistants (PDAs), and laptops. Typically, multimedia content delivery requires a large amount of transmission bandwidth. This places demands on network bandwidth capacity, which will only increase as users demand more content with higher resolution and better quality.

Due to the large bandwidth requirements and the elaborate nature of the services, the distribution and delivery of multimedia services over wireline and wireless networks presents considerable technical challenges. Modern cellular and broadband wireless technologies have the potential capabilities of delivering sufficient bandwidth to the user to support multimedia content. In order to make mobile multimedia services a reality and drive them to the mass market, a variety of technologies is being developed to create the service infrastructure to support these services within the cellular and broadband wireless infrastructure.

Multimedia services may be delivered by either unicast or broadcast/multicast transport technologies.

In certain circumstances, multiple users may be interested in the same multimedia content. To alleviate the need for data replication, some networks provide a broadcast or multicast channel that reduces the demands on the bandwidth capacity by allowing more than one user to receive the information without requiring more than one point-to-point transmission. In the 3G cellular arena, for example, protocols and frameworks have been proposed that provide Over-The-Air (OTA) broadcast physical channels in the existing cellular network. These channels can be accessed by any number of users simultaneously to receive the same content without adversely affecting bandwidth usage. Two popular cellular technologies, Global System for Mobile Communications (GSM) and Code Division Multiple Access (CDMA) both have similar capabilities. GSM, whose specifications are discussed and standardized in 3GPP (such specifications are made available at <http://www.3gpp.org>), has a service called Multimedia Broadcast/Multicast Service (MBMS). Similarly, CDMA, whose specifications are discussed and standardized in 3GPP2 (such specifications are made available at <http://www.3gpp2.org>), has a service called Broadcast-Multicast Services (BCMCS). MBMS and BCMCS are also known as in-band

2

broadcast/multicast technologies, since they operate in the same band of conventional unicast GSM (for MBMS) or CDMA (for BCMCS) services and are implemented on the same cellular networks as GSM or CDMA, just by adding additional capabilities to the existing air interfaces. The notion of in-band broadcast/multicast is not limited to cellular networks, but is also known in several broadband wireless technologies as well. For example, Worldwide Interoperability for Microwave Access (WiMAX), whose specifications are made available at <http://www.wimaxforum.org>, has a service called Multicast Broadcast Service (MCBCS) that offers these capabilities.

Referring to FIG. 1, from a high-level architectural perspective **100**, these wireless networks include a number of similarly functional components. Some of these functional components may be implemented using more than one physical component.

A Content Provider System **102** is responsible for procuring and delivering the content to a Broadcast-Capable Distribution Network **104**. Content is produced by content providers such as nationwide or local TV channels (not shown), and stored on servers in the network (not shown) for subsequent retrieval and distribution. Different content providers may use different servers to store their own content. Typically, the network operator puts in place commercial agreements with one or more content providers. Once the agreements are in place, the network operator retrieves the content and prepares it for distribution. The Content Provider System **102** usually has an interface **106** with a Controller **108** and a data path **110** with the Broadcast-capable Distribution Network **104**.

The Broadcast-Capable Distribution Network **104** (BCDN) receives content from the Content Provider System **102** and is responsible for all processing needed to transmit the content over-the-air to all the users authorized to receive content in that network. The specific processing done depends on the particular technology being used. The BCDN network **104** has the capability of broadcasting or multicasting the content using one of the technologies described above. From a functional point of view, this component is typically responsible for setting up the bearer path in a Radio Access Network (RAN) (not shown). Conceptually, the bearer path is the unicast or multicast connection through which the content is delivered to the users. This functionality is achieved in one or more of the nodes in the network.

A Broadcast-Capable Air Interface **112** is dependent on the particular technology being used. For the air interface to be broadcast capable, it has to provide capabilities for creating a multi-access Over-the-Air (OTA) channel, meaning that multiple users can tune to the same channel and receive the same content without interfering with one another.

One or more Compatible Handsets **114** are used by customers. For the handsets **114** to be compatible with a Broadcast-Capable Distribution Network **104**, the handsets **114** must have capabilities to access the Broadcast channels and must also be content-compatible with the Controller **108** and the Content Provider System **102**. This latter functionality typically includes means to signal control information to the functional blocks in the network, and to decrypt the content being transmitted. Typically, the handsets **114** also have the ability to access conventional cellular channels.

The Controller **108** is a central network functionality that interacts with the Network Operator to receive programming information and then disseminates this information to the Content Distribution System and the Network Processing Framework in the Broadcast-Capable Distribution Network

104 for the proper set up of the bearers. It also generates and distributes the Program Guide through which the users can select the content that they want to consume. It is responsible for authenticating users as well as for providing them with any information needed to decrypt the content. Additionally, the Controller **108** participates in the collection of records for billing purposes.

An Authenticate, Authorize, and Accounting (AAA) framework **116** is owned by the Network Operator and stores user information like service profiles and authentication data.

A Charging and Billing System **118** is used by the Network Operator to collect usage information so that the User can be correctly billed.

In addition to the in-band broadcast/multicast technologies listed above, some other OTA multicast technologies have been proposed. These technologies are known as out-of-band multicast technologies, and require a wireless network separate from the cellular network and typically are dedicated to the distribution of multicast content. These technologies include MediaFLO, whose specifications are made available at <http://www.floforum.org>, and Digital Video Broadcasting—Handheld (DVB-H), whose specifications are made available at <http://www.dvb.org>. These multimedia distribution systems send content to a user device using dedicated spectrums and separate physical interfaces and networks. Unlike the cellular networks, these dedicated networks are often only able to distribute content from the network to the user. Also, these technologies typically use a high level architecture that includes similarly functional elements as those for the in-band multicast technologies described above.

While multicast is well suited for some multimedia services, there are many other services that are best provided with a unicast transmission to a user. These include network-mobile services like video-on-demand (VOD) as well as mobile-mobile services like Video Telephony. Frameworks and technologies have been developed to support, create, and manage unicast multimedia services.

Once such framework is the IP Multimedia Subsystem (IMS) network architecture that specifies standard SIP and Diameter-based Interfaces between functional entities like Application Servers, Call/Service Control Functions, Home Subscriber Servers, IMS Terminals etc. The specification of the SIP and Diameter protocols are made available at <http://www.ietf.org> and comprise a number of specifications that include IETF RFC 3261, IETF RFC 2327, and IETF RFC 3588. The IMS network architecture is defined by “IP Multimedia Subsystem, Stage 2, 3GPP TS 23.228,” and other specifications available at <http://www.3gpp.org>, and is incorporated in this description by reference. The IMS architecture has also been defined by 3GPP2 in “All-IP Core Network Multimedia Domain, 3GPP2 X.S0013-000-0” and other specifications available at <http://www.3gpp2.org>. Within the 3GPP2 standard specifications, the IMS architecture is referred to as Multimedia Domain (MMD) architecture.

Referring to FIG. 2, an IMS network **200** (“3GPP IMS network”) implemented in accordance with the IMS network architecture as defined by “IP Multimedia Subsystem, Stage 2, 3GPP TS 23.228,” employs a peer-to-peer model of communication, which is suitable for unicasting information. The 3GPP IMS network **200** of FIG. 2 includes a number of functional components each configured to interoperate in the IMS network architecture as defined by “IP Multimedia Subsystem, Stage 2, 3GPP TS 23.228.” The functional components of the 3GPP IMS network **200**

include a home subscriber server **202** (HSS), a subscriber location function **204** (SLF), call session control functions **206** (CSCF), an IMS application server **208** (AS), and a media resource function (MRF). Other functional components (not shown) can be included in the 3GPP IMS network **200**. Each functional component can be implemented in one or more network entities (i.e., physical nodes) of the 3GPP IMS network **200**. Each 3GPP IMS network entity can implement one or more functional components. The IMS network architecture also defines other components (in addition to the ones listed above) which are not described in this description.

The HSS **202** is a database that contains the user profile data, such as subscription-related information, and performs authentication, authorization, and accounting functions.

The SLF **204** is an optional functional component of the 3GPP IMS network **200** that is used when multiple HSSs **202** are deployed in the 3GPP IMS network **200**. Both the HSS **202** and the SLF **204** implement the Diameter Base Protocol (details of which are provided in “IETF RFC 3588,” available at <http://www.ietf.org>, and is incorporated in this description by reference). The SLF **204** includes a table that maps each IP address of a user device (illustratively depicted as an IMS-Enabled Mobile Station **210**) to the specific HSS **202** storing information for that user device.

The CSCF **206** processes Session Initiated Protocol (SIP) signaling packets in the 3GPP IMS network **200**. There are generally three types of CSCFs **206** in the 3GPP IMS network **100**, namely: (1) the Proxy-CSCF (P-CSCF), which is the first point of contact between an IMS-Enabled Mobile Station **210** and the 3GPP IMS network **200**, provides some security functions, and interacts with the HSS **120** to authenticate the IMS-Enabled Mobile Station **210** and to determine the service capabilities that the IMS-Enabled Mobile Station **210** is authorized to access; (2) the Interrogating-CSCF (I-CSCF), which serves as an intermediate SIP router; and (3) the Serving-CSCF (S-CSCF), which performs both SIP routing as well as session control.

The AS **208** hosts and executes services (e.g., push-to-talk and instant messaging), and interfaces with the S-CSCF using SIP. Depending on the array of multimedia services supported by the 3GPP IMS network **200**, there can be any number of AS deployed in the 3GPP IMS network **200**. The term “AS” **208** refers to the union of the multiple AS deployed in the 3GPP IMS network **200**.

The MRF provides media streams to the user devices. The MRF includes a controller **212** (MRFC) and a processor **214** (MRFP). The MRFC **212** works on the control plane and does media related signaling in order to initiate, manage, and terminate streams and programs. The MRFC **212** receives instructions from the AS **208** and interprets them to the MRFP **214**. The MRFP **214** works on the data plane, and provides the functionality to collect the content, to prepare it for distribution in the appropriate format that can be understood by the user devices, and to stream the multimedia content to the user devices.

The IMS Enabled Mobile Stations are SIP-capable handsets that implement IMS capabilities and can therefore communicate with a IMS network. Specifically, the handsets implement functionality like IMS Registration, communication with CSCF and AS entities etc.

SUMMARY

In one aspect, the invention features a method that includes enabling a network entity to control delivery of

5

information to user devices over bearer paths, the bearer paths comprising unicast channels and multicast channels.

Implementations of the invention may include one or more of the following.

The method of enabling may include provisioning a multimedia service so that one or more multimedia programs are available for delivery to the user devices. For each of the one or more multimedia programs, the provisioning may include receiving program-related parameters, network-related parameters, and user-related parameters from a network operator. The provisioning may include adding the one or more multimedia program to a list of multimedia programs available for delivery to the user devices.

The method of enabling may include generating a program guide that provides a list of multimedia programs available for delivery to the user devices.

The method of enabling may include sending to a distribution network control information for use in setting up the bearer paths with the user devices.

The method of enabling may include sending to at least one of the user devices a program guide that provides a list of multimedia programs available for delivery. The program guide may be sent upon registration of the at least one of the user devices with a distribution network. The program guide may be sent in response to a program guide request by the at least one of the user devices. At least one of the user devices is associated with a service profile that identifies one or more multimedia services the user device has access to.

The method of enabling may include receiving a request for a program from at least one of the user devices; and identifying a transport mode associated with the requested program, the transport mode being one of unicast or multicast. If the identified transport mode is multicast, the method of enabling may include determining whether the requested program is currently being delivered over a multicast channel to which the at least one of the user devices has access, and if so, sending information to the at least one of the user devices to enable the at least one of the user devices to extract the requested program from that multicast channel. If the identified transport mode is unicast, the method of enabling may include determining whether the requested program is currently being delivered over a unicast channel to which the at least one of the user devices has access to, and if not, sending a notification to another network entity to start delivering the requested program over a unicast channel to which the at least one of the user devices has access. The method of enabling may include retrieving a service profile associated with the at least one of the user devices; and taking an action based on the retrieved service profile. Taking an action may include one or more of the following: determining whether the at least one of the user devices is capable of accessing the requested program; and identifying a quality of service the at least one of the user devices is subscribed to receive.

The method of enabling may include maintaining a count of a number of user devices to which a program is being delivered over unicast channels; and dynamically establishing a multicast channel when the count exceeds a threshold value. The method of enabling may include sending a notification to another network entity to start delivering the program over the newly-established multicast channel; and sending information to the user devices to enable the user devices to extract the program from the newly-established multicast channel.

The bearer paths may be set up over a plurality of distribution networks, the distribution networks using different access technologies. The bearer paths may include

6

unicast channels and the distribution networks may include a Global System for Mobile Communications (GSM) network, a General Packet Radio Service (GPRS) network, a Universal Mobile Telecommunications System (UMTS), an Evolution-Data Optimized (EV-DO) network, a High-Speed Downlink Packet Access network, and a High-Speed Uplink Packet Access network. The bearer paths may include multicast channels, and the distribution networks may include a cellular multicast network that comprises a Code Division Multiple Access (CDMA) network implementing a Broadcast-Multicast Service (BCMCS) service, and a Global System for Mobile Communications (GSM) network implementing a Multimedia Broadcast/Multicast Service (MBMS) service; a broadband wireless network that comprises a Worldwide Interoperability for Microwave Access (WiMAX) network, and a Wireless Broadband (WiBro) network; a wireless multicast network that comprises a MediaFLO network, a Digital Video Broadcasting—Handheld (DVB-H) network, and a Digital Multimedia Broadcasting network (DMB); and a wired network that comprises a Digital Subscriber Line (DSL) network, a Gigabit Ethernet network, a Frame Relay network, and a cable network.

The bearer paths used to deliver a multicast service to the user devices may originate at one of the user devices.

The network entity can be configured to interoperate in one of a 3GPP Internet Protocol Multimedia Subsystem (IMS) network architecture and a 3GPP2 Multimedia Domain (MMD) network architecture. The network entity may provide functionality of a modified 3GPP2 Broadcast and Multicast Service (BCMCS) controller component configured to enable BCMCS signaling protocol transactions to occur over interfaces of one of a 3GPP IMS network architecture and a 3GPP2 MMD network architecture.

The method of enabling may include performing 3GPP2 BCMCS service discovery and announcement functionality by generating a program guide and delivering the program guide to the user devices.

The method of enabling may include performing 3GPP2 BCMCS content subscription and information functionality by authenticating and authorizing the user devices and delivering encryption keys and control information in a way compatible to 3GPP2 BCMCS transactions performed over interfaces of one of the 3GPP IMS network architecture and the 3GPP2 MMD network architecture.

The method of enabling may include performing 3GPP2 BCMCS content delivery functionality by having a MRFC functional entity of one of the 3GPP IMS network architecture and the 3GPP2 MMD network architecture send control information to a MRFP functional entity of one of the 3GPP IMS network architecture and the 3GPP2 MMD network architecture so as to enable content to be sent content to user devices through a distribution network.

The method of enabling may include performing 3GPP2 BCMCS controller discovery functionality by having a S-CSCF functional entity of one of the 3GPP IMS network architecture and the 3GPP2 MMD network architecture send user requests to a network entity.

The method of enabling may include performing 3GPP2 BCMCS accounting functionality by having an HSS functional entity of one of the 3GPP IMS network architecture and the 3GPP2 MMD network architecture collect accounting records in a manner compatible with BCMCS transactions performed over the interfaces of one of the 3GPP IMS network architecture and the 3GPP2 MMD network architecture.

The method of enabling may include performing 3GPP2 BCMCS subscription profile manager functionality by hav-

ing a network operator configure user subscription information in a HSS functional entity of one of the 3GPP IMS network architecture and the 3GPP2 MMD network architecture.

An interface between a 3GPP2 BCMCS controller and a user device may be implemented with a SIP Interface between the network entity and the user device; an interface between a 3GPP2 BCMCS controller and a broadcast serving node may be implemented with a Mb Interface; an interface between a 3GPP2 BCMCS controller and a content server may be implemented with a Mp Interface; an interface between a 3GPP2 BCMCS controller and an authenticate, authorize, and accounting (AAA) server may be implemented with a Sh Interface; and an interface between a 3GPP2 BCMCS controller and a subscription profile manager may be implemented with a Sh interface, wherein each of the SIP Interface, Mb Interface, Mp Interface, Sh Interface is an interface of one of the 3GPP IMS network architecture and the 3GPP2 MMD network architecture.

In another aspect, the invention features a method that includes enabling a network entity configured to interoperate in one of a 3GPP Internet Protocol Multimedia Subsystem (IMS) network architecture and a 3GPP2 Multimedia Domain (MMD) network architecture to provide network-mobile multimedia services to user devices, wherein content associated with the multimedia services are stored in a plurality of storage devices in the network.

Implementations of the invention may include one or more of the following.

A bearer path used to deliver a multimedia service to the user devices may be established before an explicit service request is received from a user device. The method of enabling may include receiving from at least one of the user devices a request for authorization to access at least one of the multimedia services. The method may include establishing a multicast channel of a distribution network over which the content is simultaneously delivered to the user devices. The establishing may be performed by another network entity in response to a request for the content received from at least one of the user devices.

In another aspect, the invention features a method that includes providing a common interface through which a network operator defines service-specific parameters of a plurality of unicast and multicast multimedia services deployed in a distribution network.

Implementations of the invention may include one or more of the following.

The service-specific parameters may be common to all programs associated with the multimedia service. The service-specific parameters may include one or more of the following: transport mode parameters, geographical area parameters, encryption mode parameters, quality of service parameters, network configuration parameters, security and access parameters, announcement parameters, and charging model parameters.

Through the common interface, the network operator may define program-specific parameters of a program associated with the multimedia service. The program-specific parameters may include one or more of the following: a program name, a program identifier, a program availability time, a program start time, a program end time, transport mode parameters, encryption mode parameters, quality of service parameters, application bit rate parameters, and user information that specifies a service profile associated with the program. One or more of the program-specific parameters may override corresponding one or more of the service-specific parameters.

Through the interface, the network operator may define information to generate a program guide that provides a list of programs available for delivery to user devices in communication with the distribution network.

Through the common interface, the network operator may define rendering parameters of a program guide to be displayed on user devices in communication with the distribution network, parameters associated with program guide delivery, and parameters associated with program guide geography.

The common interface may reside on a network entity that is configured to interoperate in a 3GPP Internet Protocol Multimedia Subsystem (IMS) network architecture or a 3GPP2 Multimedia Domain (MMD) network architecture.

In other aspects, corresponding computer programs and apparatus are also provided.

Advantages that can be seen in particular implementations of the invention include one or more of the following.

A new IMS functionality that combines IMS Application Server and Media Resource Function Controller functionality is provided on a multimedia controller. The multimedia controller delivers unicast and multicast services via multiple transport technologies including cellular unicast technologies (e.g., GSM, GPRS, UMTS, 1xEVDO), cellular multicast technologies (e.g., BCMCS, MBMS), broadband wireless technologies (e.g., WiMAX, WiBro), wireless multicast technologies (e.g., MediaFlo, DVB-H), and wired technologies (e.g., DSL, cable). The multimedia controller enables a dynamic switchover between unicast and multicast delivery modes, based on the number of users that request delivery of a particular program.

Although described in the context of an enhanced IMS network, the multimedia controller can be implemented in any number of network architectures to provide both unicast and multicast multimedia services over said networks.

The enhanced IMS network includes the capability to support a network-device model, which is used to enable delivery of network-generated multicast and unicast content to user devices.

The Service and Program Design Interface enables a network operator to configure service parameters and programming schedules, and easily and efficiently disseminate the programming schedules to any network in which unicast and multicast multimedia services are supported. In those instances in which the Service and Program Design Interface is provided in the context of an enhanced IMS network, the interface enables the network operator to easily and efficiently convey program schedules to the multimedia controller for further dissemination to the distribution network and the content provider system.

The modified architecture for BCMCS implements the BCMCS protocols over interfaces of one of a 3GPP IMS network architecture and a 3GPP2 MMD network architecture. This allows the network operator to integrate BCMCS into their 3GPP IMS and/or 3GPP2 MMD architectures. To this end, the BCMCS interactions are modified to take place over the IMS/MMD Mp and Cx Interfaces. The role of the BCMCS Controller is redefined as a combined IMS/MMD Application Server and Media Resource Function Controller, called the multimedia (BCMCS) controller. This multimedia (BCMCS) controller is designed to serve multimedia to the users using both Multicast (BCMCS) as well as Unicast (regular 1xEVDO).

Particular implementations of the invention also provide network operators and users with the following IMS-related advantages: (1) it simplifies introduction of new unicast and multicast multimedia services; (2) it provides a standard

architecture to create and maintain different kinds of unicast and multicast multimedia services; (3) it presents a common environment to a user to access multiple unicast and multicast services; and (4) it provides an access-agnostic way to a network operator to disseminate unicast and multicast multimedia content.

The details of one or more examples are set forth in the accompanying drawings and the description below. Further features, aspects, and advantages of the invention will become apparent from the description, the drawings, and the claims.

DESCRIPTION OF DRAWINGS

FIG. 1 is a block diagram of a broadcast-capable network.
FIG. 2 is a block diagram of a 3GPP IMS Network.
FIG. 3 is a block diagram of an enhanced IMS Network.
FIGS. 4-6 each shows a call flow.
FIG. 7 is a block diagram of a 3GPP2 BCMCS Network.
FIG. 8 is a block diagram of an adaptation of BCMCS to IMS.

DETAILED DESCRIPTION

Multimedia Controller

FIG. 3 shows an example network (“enhanced IMS network”) 300 in which multimedia services generated by content providers 302 can be delivered to user devices (depicted as IMS-enabled mobile stations 304) through bearer paths of a distribution network 306. The bearer paths include unicast channels and multicast channels, and the distribution network 306 includes one or more of the following networks: a Global System for Mobile Communications (GSM) network, a General Packet Radio Service (GPRS) network, a Universal Mobile Telecommunications System (UMTS), an Evolution-Data Optimized (EV-DO) network, a High-Speed Downlink Packet Access network, a High-Speed Uplink Packet Access network, a Code Division Multiple Access (CDMA) network implementing a Broadcast Multicast Service (BCMCS) service, a GSM network implementing a Multimedia Broadcast/Multicast Service (MBMS) service, a Worldwide Interoperability for Microwave Access (WiMAX) network, a Wireless Broadband (WiBro) network, a MediaFLO network, a Digital Video Broadcasting—Handheld (DVB-H) network, a Digital Multimedia Broadcasting network (DMB), a Digital Subscriber Line (DSL) network, a Gigabit Ethernet network, a Frame Relay network, and a cable network.

The enhanced IMS network 300 includes a functional entity referred to in this description as a “multimedia controller” 308 that implements two functions, namely: (1) multimedia service and program provisioning; and (2) multimedia program delivery. Other functions may also be implemented by the multimedia controller 308.

Although described in the context of an enhanced IMS network, the Multimedia Controller can be implemented in any number of network architectures to provide both unicast and multicast multimedia services over said networks.

Multimedia Service and Program Provisioning

The multimedia controller 308 provides an interface through which an operator of the enhanced IMS network 300 provisions multimedia services and programs on a recurring basis (e.g., weekly) or as needed (e.g., when new programs need to be provisioned). In one example, the

interface provided by the multimedia controller 308 is a “Service and Program Design Interface” 310 (details of which are described below) that runs on a server in the enhanced IMS network 300. Referring also to the call flow of FIG. 4, for each program, the network operator enters a number of program-related parameters, an illustrative list of which is included below:

Program Name: Text string describing the name of the program

Program ID: An identifier which is used in communications between the AS and other entities

Program Availability Time: When the program is available in the enhanced IMS network

Program Start & End Time: Only applicable to programs to be multicast

Encryption details: Specifies which encryption mode is used and associated information

Header Compression details: Specifies header compression parameters

Application Bit Rate: The rate at which the content is to be sent

Quality of service (QoS) Parameters: If the program has any QoS requirements

Transport mode: Whether this program is to be streamed over a unicast channel, a multicast channel, or both

Transport information: Based on the transport mode, this includes extra information like IP addresses, ports, geographical information etc.

User information: Specifies service profile associated with this program

Once the network operator has entered the relevant parameters, the interface 310 initiates communication with the AS-portion 312 of the multimedia controller 308 and the HSS 202, and delivers the appropriate information to each. In one example, the interface 310 sends the HSS 202 the user-profile information to enable the HSS 202 to determine whether a particular user is allowed to access the program, and sends the rest of the information to the AS-portion 312 of the multimedia controller 308. The AS-portion 312 of the multimedia controller 308 sends the information it receives to the MRFC 314 to enable the MRFC 314 to generate or update a program guide that provides a time-sensitive listing of all multimedia programs that are available for delivery to the user devices. The MRFC 314 sends the program guide through the standard IMS Mp interface to the MRFP 316 at the multimedia content server, which in turn sends the program guide to the distribution network 306 for dissemination at regular intervals to the users. The MRFP 318 part of the multimedia controller 308 also uses the Mb interface to communicate program scheduling details to the distribution network 306 for use in establishing the bearer paths at the appropriate times (e.g., in the case of Network-Initiated broadcast).

The program guide can be delivered to a user device upon IMS Registration. As part of standard IMS operation, initially, the user device performs the IMS Registration process with the S-CSCF 320. This involves the S-CSCF 320 verifying the identity of the user device with the HSS 202 and downloading some user information (including a service profile) from the HSS 202. If the service profile indicates that the user can access multimedia programs, the program guide is automatically delivered to the user device. Once rendered on a screen of the user device, a user of the user device can request the delivery of a multimedia program by selecting one of the multimedia programs listed in the program guide. This request is routed through the distribution network to the multimedia controller 308, which then

11

takes either of the following actions based on the transport mode of the selected multimedia program:

Unicast mode: The multimedia controller **308** passes on the IP Address of the user device to the MRFP **316** at the multimedia content server so that it can send the requested multimedia program directly to the user device. One example call flow in the context of a 1xEVDO distribution network is shown in FIG. **5**. In this example, the MRFP **316** at the multimedia content server sends the content to the user device via a packet data serving node (PDSN) and a radio network controller (RNC).

Multicast mode: The multimedia controller **308** sends information to the user device sufficient to enable the user device to extract the multimedia program from a multicast channel. One example call flow in the context of a 1xEVDO distribution network implementing a BCMCS service is shown in FIG. **6**. In this example, the user device requires encryption keys, header compression parameters, and BCMCS FlowID to Multicast IP Address mappings. Additional information concerning the BCMCS service is described in U.S. application Ser. No. 11/229,180 filed on Sep. 15, 2005, and incorporated by reference.

The program guide can also be delivered to a user device in response to a specific program guide request received from a user device, or as part of a periodic broadcast to update user devices with the latest program guide. When rendered on a user device, the program guide provides the list of multimedia programs that are available for selection by the user device. This list includes all programs that have been provisioned regardless of the transport mode (i.e., the list includes programs to be unicasted and programs to be multicasted).

In those instances in which the HSS **202** is required to act as a key generator (e.g., BCMCS requires periodically-updated Broadcast Access Key and Short-Term Keys), the AS **312** subscribes to an event-notification feature of the HSS **202** so as to be notified when the key(s) change. When the AS **312** receives the new keys, the AS **312** conveys them to the MRFC **314** and the MRFP **318** so that the multimedia program can be accurately encrypted.

The multimedia controller **308** registers for IMS event notification with the S-CSCF **320**. This action ensures that the multimedia controller **308** is notified whenever a user device, whose service profile includes access to multimedia services/programs, registers with the S-CSCF **320**.

In other examples, the program guide is generated by an AS other than the AS-portion **312** of the multimedia controller, and the functionality to render the program guide is provided by the MRFP **316** at the multimedia content server. Although the description above provides for a user selection of a multimedia program by way of a program guide, in other examples, the user selects the program through other means, for example, by accessing a known website.

Multimedia Content Delivery

Multimedia content can be delivered to user devices using either or both of the following transport modes:

Unicast streaming: the program is sent over a unicast channel to an individual user device. Typically, the program is delivered in response to a specific request received from the user device.

Multicast streaming: the program is sent over a broadcast channel and can be accessed by a large number of user devices simultaneously. In some instances, the program

12

is delivered regardless of the presence of user devices in a sector. In some instances, the program is delivered on demand based on user activity.

The multimedia controller **308** and the user devices are aware of the transport mode with which the program is delivered. However, users of the respective user devices are unaware of the actual transport mode of the multimedia program, and do not experience any change in user experience based on the transport mode. That is, the same user experience is achieved regardless of the transport mode.

FIGS. **5** and **6** show example call flows associated with multimedia content delivery. The sequence of events includes:

1. The user device already has a program guide, which lists all multimedia programs available for delivery in the geographic area in which the user device is located. The term "geographical area" indicates a terrestrial region—this could be a cell sector, a combination of many sectors, a city, a state or a country.
2. The user of the user device selects a program from the program guide.
3. This request is conveyed to the AS-portion **312** of the multimedia controller by the CSCFs **320**.
4. The multimedia controller **308** uses the IMS Sh Interface to communicate with the HSS **202** to determine if the user device can access the selected program. The multimedia controller **308** also retrieves the service profile associated with the user device which indicates, among other things, the level of QoS that the user device is subscribed to receive.
5. If the transport mode of the selected program is designated as unicast; the multimedia controller sends the IP address of the user device, program rate and QoS settings to the MRFC **314**, which passes them on to the MRFP **318** at the multimedia content server.
6. The MRFP **316** at the multimedia content server delivers the selected program to the user device over a unicast channel of the distribution network.
7. If the transport mode of the selected program is designated as multicast and is currently being multicasted, the AS-portion **312** of the multimedia controller simply conveys some information (in the case of BCMCS, the information includes encryption keys, header compression parameters, and BCMCS FlowID to Multicast IP Address mappings) to the user device to enable the user device to extract the program from the appropriate multicast channel of the distribution network and decrypt the extracted program.
8. If the transport mode of the selected program is designated as multicast, is not currently being multicasted, and the enhanced IMS network permits user-initiated multicasting, the AS-portion **312** of the multimedia controller establishes the multicast channel of the distribution network and informs the MRFP **316** at the multimedia content server to begin delivering the selected program over the newly-established multicast channel.

In some implementations, the multimedia controller **308** maintains a count of a number of user devices to which a program is being delivered over unicast channels of a single unicast network. The multimedia controller **308** dynamically establishes a multicast channel when the count exceeds a predefined threshold value, sends a notification to the MRFP **316** at the multimedia content server to start delivering the program over the newly-established multicast channel, and sends information to each of the user devices to enable the user devices to extract the program from the newly-established

lished multicast channel. In so doing, if more user devices request the same program, the enhanced IMS network **300** can deliver the program in a more efficient manner to reduce the demands on the network bandwidth capacity

Network-Device Model

As previously discussed, the IMS network architecture as currently defined by "IP Multimedia Subsystem, Stage 2, 3GPP TS 23.228," implements a peer-to-peer model. In a peer-to-peer communication, a specific user device explicitly initiates a communication with another user device by requesting the IMS network to establish a point-to-point connection. In such instances, the 3GPP IMS network creates connection between the two user devices, and one of the user devices provides the content or media that is sent to the other of the user devices.

The enhanced IMS network **300** is implemented to support a network-device model that enables the network **300** to deliver network-provided multicast content to the handsets. In such instances, the network **300** may establish the bearer path without an explicit request from a user device, or the bearer path may be established upon a request by the user device. After the bearer path is created, a user device accesses the bearer path to receive the associated content. Typically, in order to access the bearer path, the user device requests certain functional entities in the network **300** for authentication and access authorization. The bearer path can also be created as a response to a request by a user device for a specific program. In this case, depending on network operator provisioning, the bearer path may be unicast or multicast. If, after a unicast bearer has been created, other user devices request the same program, the network **300** is implemented with functionality that supports a dynamic switchover from a unicast mode of delivery to a multicast mode of delivery, as the latter is more efficient.

The content to be multicasted can be a multimedia program (e.g., a television program, a video program, a radio program, a music distribution service, a pod cast, a program guide that provides a listing of multimedia programs available for delivery to the user devices, an advertisement, a network-related message, or an emergency alert message) accessible by the MRFP **316** at the multimedia content server.

The enhanced IMS network **300** supports all multimedia services, both unicast and multicast, and inherits the benefits of the 3GPP IMS framework including rapid creation of new services, and unified management of all services.

Although the enhanced IMS network **300** provides the content, the content may originate (i.e., be generated) at a user device. Since the user terminal does not have the capability to multicast the content, multicasting can be achieved by sending the content to a server in the enhanced IMS network **300**, which in turn provides it to a desired group of user devices. Two example services in which such techniques would be useful is a push-to-talk service with multicast delivery and a push-to-media service with multicast delivery.

Service and Program Design Interface

In one implementation, the multimedia controller **308** provides a Service and Program Design Interface **310** that enables a network operator to easily and efficiently define each multimedia service supported by the enhanced IMS network **300**, and define each program associated with a multimedia service supported by the enhanced IMS network

300. Once the multimedia services and programs are defined, the interface **310** is configured to generate a program guide which is delivered to the user devices and used by the user to select the programs, and also contains information that is used in establishing bearer paths in the distribution network **306**.

Through the interface **310**, the network operator can define parameters that are common to all programs that are associated with a specific multimedia service. Examples of such service-related parameters include:

Transport parameters: the way the service is delivered (multicast, unicast, or a combination of unicast and multicast), encoding/encryption specifications, Quality of Service requirements, etc.

Network configuration parameters: based on the transport mode, these network configuration parameters provide extra information such as IP addresses, ports, geographical information, etc.

Security and access parameters

Announcement parameters: how the program guide is to be constructed, where the program guide is to be located, how frequently the program guide is updated, how the program guide is delivered, etc.

Charging and billing parameters: how the network operator collects usage information so that the user device can be correctly billed

Through the interface **310**, the network operator can also define parameters that are specific to a particular program. Examples of such program-related parameters include a program name, a program identifier, a program availability time, a program start time, a program end time, transport mode parameters, encryption mode parameters, quality of service parameters, application bit rate parameters, and user information that specifies a service profile associated with the program.

Generally, in those instances in which the program-related parameters conflict with the service-related parameters, the program-related parameters take precedence and override the service-related parameters.

In addition, the interface **310** enables a network operator to easily and efficiently set up program schedules (for example details like Program Name, Time, Bit Rate etc) to the multimedia controller **308** for further dissemination on to the distribution network **306** and the content provider system **302**.

The following is an example scenario in which the network operator interacts with the Service and Program Design Interface **310** to define a service:

1. Define program guide rendering. Parameters that are defined in this step include:

The geometry of how the program guide is rendered, such as how channels are displayed and how the time grid is rendered.

Common elements to be rendered in the program guide, such as channel logos.

Where the information needed to build the program guide is stored, how that information is read, and whether a parser is used to read that information.

The exact behavior of the program guide during channel selection (e.g., the selected program gets highlighted)

Other rendering information, which may be customized to each type of mobile device supported by the network.

The program guide may also include other services such ticker services, which are also defined in this step.

2. Define program guide delivery method, such as unicast, multicast, or a combination of the two. Certain portions of the program guide may be delivered only once using unicast or multicast (for example, common elements that are not time sensitive), while other elements may be delivered periodically (in the latter case, define the period of time).
3. Define program guide geography. Certain selections may be displayed only in certain geographical areas.
4. Define service access/security. Access to content may depend on the type of subscription, or on type of content. Premium and basic packages of program may be defined, each corresponding to different access rights. The composition of each program package may depend on geography.
5. Define billing and charging models. Possible models include: flat rate, pay-per-volume, pay-per-view, pay-per-program, etc.

Once a service is defined, individual programs associated with the service can be provisioned, and subsequently delivered to the user devices. The provisioning of each individual program includes the program-specific parameters described above, and a specification on how the enhanced IMS network **300** is to handle errors.

Although described in the context of an enhanced IMS network **300**, the Service and Program Design Interface **310** can be implemented in any number of network architectures and provided by any number of different network components.

Modified BCMCS Controller

3GPP2 has produced a technical specification "Broadcast and Multicast Service in cdma2000 Wireless Network," 3GPP2 X.S0022-0, available at <http://www.3gpp2.org>, that described a framework called Broadcast-Multicast Services (BCMCS) over a 1xEVDO network. This technical specification describes the interfaces between access terminals, access network, external entities such as a content server, an AAA (authentication, authorization, and accounting) server, a subscription profile manager, and a BCMCS controller.

Referring to FIG. 7, the access terminals **702** are 1xEVDO handsets that are BCMCS-capable. These mobile stations **702** have the ability to access BCMCS channels in the access network **704**, and typically include the ability to play audio/video streams on a viewer of the access terminal **702**. The access network **704** is a 1xEVDO Radio Access Network augmented with BCMCS capabilities. The access network **704** provides the access terminals **702** with IP connectivity and provides the bearer paths for the BCMCS content packets. The access network **704** typically includes a broadcast serving node **706** that implements various BCMCS protocols (e.g., Broadcast Framing Protocol and Broadcast MAC Protocol), and is responsible for establishing the multicast bearer path in the access network **704**. The content server **708** interfaces with the broadcast serving node **706** to provide content streams. The subscription profile manager **710** manages a subscription profile database, which contains user profiles for the subscribers. The BCMCS controller **712** implements a number of different functions including the following:

It provides the mobile stations with program information, encryption keys, header compression parameters and any other data the mobile stations need to recover the higher-layer packets.

It provides the broadcast serving node with the program schedule.

It provides the content server with the program schedule.

It interacts with the AAA server **714** to authenticate the user device and to upload billing records.

The 3GPP2 BCMCS framework enables a network operator to provide multimedia-broadcasting services using BCMCS-capable access networks and mobile stations. However, the 3GPP2 BCMCS framework is incompatible with the IMS network architecture defined by "IP Multimedia Subsystem, Stage 2, 3GPP TS 23.228."

To address this incompatibility, this description provides a modified architecture for the BCMCS controller in which BCMCS signaling protocol transactions over IMS interfaces are defined. Such a modified BCMCS controller enables the network operator to integrate BCMCS functionality into its existing IMS network. To this end, the BCMCS signaling protocol transactions are modified to take place over the IMS Mp and Cx interfaces, rather than the current RADIUS, XML interfaces.

In the implementation of an IMS network **800** depicted in FIG. 8, a single network entity referred to in this description as a "multimedia (BCMCS) controller" **808** includes the following functionality: the AS **812**, the MRFC **814** (which corresponds to the 3GPP2 BCMCS Controller's function to convey program information to the MRFP **816** at the multimedia (BCMCS) content server), and some of the MRFP **818** functionality (which corresponds to the 3GPP2 BCMCS controller's function to provide the broadcast serving node with the program schedule so that the distribution network can set up the bearer path).

The multimedia (BCMCS) controller **808** controls the delivery of multimedia programs to user devices (illustratively depicted as IMS-Enabled Mobile Stations **804**) through multicast channels (using the BCMCS techniques (described in U.S. application Ser. No. 11/229,180 filed on Sep. 15, 2005, and incorporated by reference) over the IMS interfaces) and through unicast channels (using regular 1xEVDO techniques). As all IMS interfaces are either SIP or Diameter-based, the Controller-CS and the Controller-BSN Interactions are now are part of the SIP-based Mp Interface and the Mb Interface respectively, and the Controller-AAA Interaction is now part of the Diameter-based Sh Interface. A description of BCMCS functionalities and their IMS equivalents is shown in Table 1.

TABLE 1

BCMCS Functionality	Equivalent IMS Functionality
Service Discovery/Announcement	Program Guide Generation and Delivery to MS by MC (AS, MRF)
Content Subscriptions/Information Acquisition	MS interacts with MC (AS) to authenticate and authorize itself and receive encryption keys, mapping, etc.
Content Availability Determination	Handled by Access Network
Content Delivery	Interaction between MC (MRFC) and CS (MRFP) to send Content to BSN
Controller Discovery	S-CSCF routes MS's request to MC
Bearer Path Setup	Interaction between MC (MRFP) and BSN to set up bearer path
Accounting	Collection of records by AS, MRFP and upload of records to HSS
Subscription Profile Manager	Operator configures User subscription details on HSS; AS retrieves as needed

A description of BCMCS Interfaces and their Equivalent IMS Interfaces is shown in Table 2.

TABLE 2

BCMCS Interface	Equivalent IMS Interface
Controller-MS	SIP Interface between MS (SIP UAC) and MC (SIP UAS)
Controller-BSN	IMS Mb Interface
Controller-CS	IMS Mp Interface
Controller-AAA Server	IMS Sh Interface
Controller-Subscription	IMS Sh Interface
Prof. Mgr	
AN-MS	Handled by Access Network

The techniques described herein can be implemented in digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. The techniques can be implemented as a computer program product, i.e., a computer program tangibly embodied in an information carrier, e.g., in a machine-readable storage device or in a propagated signal, for execution by, or to control the operation of, data processing apparatus, e.g., a programmable processor, a computer, or multiple computers. A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program can be deployed to be executed on one computer or on multiple computers at one site or distributed across multiple sites and interconnected by a communication network.

Method steps of the techniques described herein can be performed by one or more programmable processors executing a computer program to perform functions of the invention by operating on input data and generating output. Method steps can also be performed by, and apparatus of the invention can be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application-specific integrated circuit). Modules can refer to portions of the computer program and/or the processor/special circuitry that implements that functionality.

Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. The essential elements of a computer are a processor for executing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks. Information carriers suitable for embodying computer program instructions and data include all forms of non-volatile memory, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in special purpose logic circuitry.

To provide for interaction with a user, the techniques described herein can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube) or LCD

(liquid crystal display) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer (e.g., interact with a user interface element, for example, by clicking a button on such a pointing device). Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input.

The techniques described herein can be implemented in a distributed computing system that includes a back-end component, e.g., as a data server, and/or a middleware component, e.g., an application server, and/or a front-end component, e.g., a client computer having a graphical user interface and/or a Web browser through which a user can interact with an implementation of the invention, or any combination of such back-end, middleware, or front-end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network ("LAN") and a wide area network ("WAN"), e.g., the Internet, and include both wired and wireless networks.

The computing system can include clients and servers. A client and server are generally remote from each other and typically interact over a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

Other embodiments are within the scope of the following claims. The following are examples for illustration only and not to limit the alternatives in any way. The techniques described herein can be performed in a different order and still achieve desirable results.

What is claimed is:

1. A method performed by a network entity, comprising: generating a program guide that provides a list of multimedia programs available for delivery to a user device, the program guide including a first portion comprising at least one service-specific parameter and a second portion comprising at least one program-specific parameter, wherein the at least one service-specific parameter is common to a plurality of programs in the list of multimedia programs; and sending, from the network entity to the user device, the second portion of the program guide without sending the first portion of the program guide.
2. The method of claim 1, wherein the at least one program-specific parameter is associated with at least one program in the list of multimedia programs.
3. The method of claim 1, wherein the at least one service-specific parameter includes one or more of the following: transport mode parameters, geographical area parameters, encryption mode parameters, quality of service parameters, network configuration parameters, security and access parameters, announcement parameters, and charging model parameters.
4. The method of claim 1, wherein the at least one program-specific parameter includes one or more of the following: a program name, a program identifier, a program availability time, a program start time, a program end time, encryption mode parameters, quality of service parameters, application bit rate parameters, and user information that specifies a service profile associated with the program.

19

5. The method of claim 1, further comprising, identifying a transport mode associated with a multimedia program, the transport mode indicating whether the user device will receive the multimedia program over a unicast channel or a multicast channel.

6. The method of claim 5, wherein the transport mode associated with the multimedia program is identified in accordance with at least one of the service-specific parameter and the program-specific parameter.

7. The method of claim 6, wherein one or more of the program-specific parameters overrides a corresponding one or more of the service-specific parameters.

8. The method of claim 5, further comprising:
receiving, from the user device, a request for delivery of the multimedia program; and

transmitting a communication that causes the user device to receive the multimedia program over the identified transport mode.

9. The method of claim 1, further comprising, sending the first portion of the program guide in response to the user device registering with a network.

10. The method of claim 1, further comprising, sending the first portion of the program guide in response to receiving a program guide request from the user device.

11. The method of claim 1, wherein the second portion of the program guide is sent, from the network entity to the user device, on a periodic basis.

12. A network entity comprising a memory and processing circuitry configured to:

generate a program guide that provides a list of multimedia programs available for delivery to a user device, the program guide including a first portion comprising at least one service-specific parameter and a second portion comprising at least one program-specific parameter, wherein the at least one service-specific parameter is common to a plurality of programs in the list of multimedia programs; and

send, from the network entity to the user device, the second portion of the program guide without sending the first portion of the program guide.

13. The network entity of claim 12, wherein the at least one program-specific parameter is associated with at least one program in the list of multimedia programs.

20

14. The network entity of claim 12, wherein the at least one service-specific parameter includes one or more of the following: transport mode parameters, geographical area parameters, encryption mode parameters, quality of service parameters, network configuration parameters, security and access parameters, announcement parameters, and charging model parameters.

15. The network entity of claim 12, wherein the at least one program-specific parameter includes one or more of the following: a program name, a program identifier, a program availability time, a program start time, a program end time, encryption mode parameters, quality of service parameters, application bit rate parameters, and user information that specifies a service profile associated with the program.

16. The network entity of claim 12, further configured to identify a transport mode associated with a multimedia program, the transport mode indicating whether the user device will receive the multimedia program over a unicast channel or a multicast channel.

17. The network entity of claim 16, wherein the transport mode associated with the multimedia program is identified in accordance with at least one of the service-specific parameter and the program-specific parameter.

18. The network entity of claim 17, wherein one or more of the program-specific parameters overrides a corresponding one or more of the service-specific parameters.

19. The network entity of claim 16, further configured to: receive, from the user device, a request for delivery of the multimedia program; and

transmit a communication that causes the user device to receive the multimedia program over the identified transport mode.

20. The network entity of claim 12, further configured to send the first portion of the program guide in response to the user device registering with a network.

21. The network entity of claim 12, further configured to send the first portion of the program guide in response to receiving a program guide request from the user device.

22. The network entity of claim 12, wherein the second portion of the program guide is sent, from the network entity to the user device, on a periodic basis.

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